ChatGPT - nlp u4

> i will give my textbook notes explain all in detail of natural language processing

Absolutely. Send me your textbook notes (text, images, or scanned pages), and I'll explain them in detail.

For each topic, I can provide:

1. **Simple explanation** (beginner-friendly)
2. **Detailed technical explanation**
3. **Definitions of important terms**
4. **Step-by-step examples**
5. **Diagrams/tables (when useful)**
6. **Mathematical formulas and derivations**
7. **Exam-oriented notes**
8. **Short summary for revision**
9. **Possible viva/interview questions**
10. **Practice questions with answers**

For NLP topics, I can explain concepts such as:

- Tokenization
- Stemming & Lemmatization
- N-grams
- Part-of-Speech Tagging
- Parsing
- Named Entity Recognition (NER)
- Language Models
- Word Embeddings (Word2Vec, GloVe, FastText)
- RNN, LSTM, GRU
- Attention Mechanism
- Transformers
- BERT, GPT
- Machine Translation
- Sentiment Analysis
- Information Retrieval
- Text Classification
- Speech Processing

- NLP Evaluation Metrics (BLEU, ROUGE, Perplexity, etc.)

Just paste the notes or upload images/PDF pages, and I'll explain every line in a detailed NLP-focused manner.

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Chapter 1: Hello Transformers — Detailed NLP Explanation

This chapter introduces the foundations of modern **Natural Language Processing (NLP)** and explains how **Transformers** revolutionized the field.

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1. Introduction to Transformers

In **2017**, researchers at Google published the paper:

"Attention Is All You Need"

This paper introduced a new neural network architecture called the **Transformer**.

Before Transformers, NLP systems mainly used:

- Recurrent Neural Networks (RNNs)
- Long Short-Term Memory Networks (LSTMs)

Transformers achieved:

✅ Better translation quality

✅ Faster training

✅ Better parallel processing

✅ Better handling of long sentences

This became the foundation for:

- GPT
- BERT

which dominate NLP today.

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Why Were Transformers Needed?

Imagine translating:

> "The boy who was standing near the large tree in the garden was reading a book."

The model must remember information from the beginning until the end.

Older RNN models struggled because:

- Information fades over long sequences.
- Training is slow.
- Computations are sequential.

Transformers solved these issues using:

1. Encoder-Decoder Architecture
2. Attention Mechanism
3. Transfer Learning

These are the three core topics of this chapter.
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2. Encoder–Decoder Framework

Before Transformers, NLP tasks like translation used an **Encoder-Decoder Architecture**.

What is an Encoder?

The encoder reads the input sentence and converts it into a numerical representation.

Example:

Input:

```
Transformers are great
```

Encoder converts it into:

```
Vector Representation
```

This vector contains the meaning of the sentence.

What is a Decoder?

The decoder takes the encoded representation and generates output.

Example:

Input English:

```
Transformers are great
```

Output German:

```
Transformer sind grossartig
```

The decoder generates one word at a time.

Workflow

Step 1

Input Sentence

```
Transformers are great
```

↓

Step 2

Encoder Processes Sentence

↓

Step 3

Creates Hidden State Vector

↓

Step 4

Decoder Reads Hidden State

↓

Step 5

Produces Translation

```
Transformer sind grossartig
```

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3. Recurrent Neural Networks (RNNs)

The encoder and decoder were often implemented using RNNs.

What is an RNN?

RNN stands for:

Recurrent Neural Network

It is designed for sequence data:

- Text
- Speech
- Time series

The special feature:

It remembers previous information.

Example

Sentence:

```
I love NLP
```

Processing:

Word 1:

```
I
```

↓

Hidden State h_1

Word 2:

```
love
```

↓

Uses h_1

↓

Creates h_2

Word 3:

```
NLP
```

↓

Uses h_2

↓

Creates h_3

Thus information flows through the sentence.

Hidden State

The hidden state is a memory vector.

It stores:

- Previous words
- Context
- Meaning learned so far

Mathematically:

$$h_t = f(x_t, h_{t-1})$$

Where:

- x_t = current word
- h_{t-1} = previous memory
- h_t = new memory

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4. Problem with Encoder-Decoder RNN

The encoder compresses the entire sentence into a single vector.

Example:

```
The weather is very beautiful today and we are planning a picnic...
```

Entire sentence →

One vector

↓

Decoder

This creates an:

Information Bottleneck

Because:

- Long sentences contain lots of information.
- A single vector cannot perfectly store everything.
- Early words may be forgotten.

Result:

Poor translations for long sentences.

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5. Attention Mechanism

Attention was invented to solve the bottleneck problem.

Main Idea

Instead of using only the final hidden state:

Decoder can access **all encoder hidden states**.

Without Attention

```
Encoder
   ↓
Single Hidden State
   ↓
Decoder
```

With Attention

```
Encoder
 ↓ ↓ ↓ ↓ ↓
All Hidden States
 ↓ ↓ ↓ ↓ ↓
Decoder
```

The decoder can now look at any part of the sentence whenever needed.

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How Attention Works

When generating a word:

The decoder asks:

> Which input words are important right now?

It assigns scores called:

Attention Weights

Example:

Input:

```
The cat sat on the mat
```

Generating:

```
cat
```

Attention might assign:

Word	Weight
The	0.05
cat	0.70
sat	0.10
on	0.05
mat	0.10

Highest attention:

```
cat
```

Therefore decoder focuses there.

Advantages of Attention

Better Long-Distance Dependencies

Example:

```
The student who won the national competition is happy.
```

Attention can directly connect:

```
student ↔ is
```

without passing through every intermediate word.

Better Translation

Different languages have different word orders.

Attention learns word alignment automatically.

Example:

English:

```
Area
```

French:

```
Zone
```

Attention learns:

```
Area ↔ Zone
```

relationship.

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6. Limitation of Attention + RNN

Although attention improved performance:

RNNs still process sequentially.

Example:

```
Word1 → Word2 → Word3 → Word4
```

Cannot process Word4 before Word3.

Therefore:

❌ Slow training

❌ Poor parallelization

❌ Difficult scaling

This motivated the Transformer architecture.

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7. Transformer Architecture

The biggest innovation:

Remove Recurrence Completely

No RNN

No LSTM

Only Attention

The Transformer uses:

Self-Attention

plus

Feed Forward Neural Networks (FFNN)

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8. Self-Attention

Normal attention:

Decoder looks at encoder states.

Self-attention:

Words look at other words in the same sentence.

Example

Sentence:

```
The animal didn't cross the street because it was tired.
```

Question:

What does "it" refer to?

Self-attention learns:

```
it → animal
```

Thus understanding context.

Why Self-Attention is Powerful

Every word can directly interact with every other word.

Example:

```
N words
```

Each word examines:

```
all N words
```

simultaneously.

Benefits:

✅ Better context

✅ Parallel computation

✅ Faster training

✅ Long-range understanding

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9. Transfer Learning in NLP

Another major breakthrough.

Before transfer learning:

For every task:

1. Build model
2. Train from scratch
3. Need lots of labeled data

Very expensive.

Transfer Learning Idea

Learn knowledge once.

Reuse it many times.

Step 1: Pretraining

Train on huge text corpus.

Examples:

- Wikipedia
- Books
- Articles

Task:

Predict next word.

Example:

```
I love eating _____
```

Predict:

```
pizza
```

This is called:

Language Modeling

The model learns:

- Grammar
- Syntax
- Semantics
- Facts

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Step 2: Domain Adaptation

Adapt pretrained model to specific domain.

Example:

General corpus:

```
Wikipedia
```

Target domain:

```
Movie Reviews
```

The model learns movie-related vocabulary.

Step 3: Fine-Tuning

Add task-specific layer.

Examples:

- Sentiment Analysis
- Spam Detection
- Question Answering

Now train using a small labeled dataset.

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10. ULMFiT

ULMFiT stands for:

Universal Language Model Fine-Tuning

It was among the first successful NLP transfer learning methods.

Process:

```
Pretraining
    ↓
Domain Adaptation
    ↓
Fine-Tuning
```

This showed NLP could benefit from transfer learning just like computer vision.

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11. GPT

GPT =

Generative Pretrained Transformer

Uses:

Transformer Decoder Only

Architecture:

```
Decoder
↓
Language Modeling
↓
Text Generation
```

Training objective:

Predict next word.

Example:

```
I love machine _____
```

Predict:

```
learning
```

Applications:

- Chatbots
- Content Generation
- Summarization
- Coding Assistants

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12. BERT

BERT =

Bidirectional Encoder Representations from Transformers

Uses:

Transformer Encoder Only

Unlike GPT:

BERT looks both directions.

Masked Language Modeling

Example:

```
I looked at my [MASK]
and saw that [MASK] was late.
```

Possible predictions:

```
watch
he
```

BERT learns by filling missing words.

Why BERT is Powerful

Understands:

- Left context
- Right context

Simultaneously.

Applications:

- Search Engines
- Question Answering
- NER
- Sentiment Analysis

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GPT vs BERT

Feature	GPT	BERT
Architecture	Decoder	Encoder
Training Objective	Next Word Prediction	Masked Word Prediction
Direction	Left → Right	Bidirectional
Best For	Generation	Understanding
Examples	ChatGPT	Search, QA

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Exam-Oriented Summary

Important Definitions

Encoder

- Converts input sequence into vector representation.

Decoder

- Generates output sequence.

RNN

- Neural network with memory for sequential data.

Attention

- Mechanism that focuses on important words.

Self-Attention

- Words attend to other words in the same sentence.

Transformer

- Architecture based entirely on attention.

Transfer Learning

- Reusing pretrained knowledge for new tasks.

ULMFiT

- First successful transfer learning framework for NLP.

GPT

- Decoder-only transformer for text generation.

BERT

- Encoder-only transformer for language understanding.

5-Mark Questions

1. Explain Encoder-Decoder Architecture.
2. What is Attention Mechanism?
3. Explain Information Bottleneck in RNNs.
4. What is Transfer Learning in NLP?
5. Compare GPT and BERT.

10-Mark Questions

1. Explain Transformer Architecture in detail.
2. Discuss Self-Attention and its advantages.
3. Explain ULMFiT and Transfer Learning.

4. Compare RNN, LSTM, and Transformer architectures.

This covers the entire Chapter 1 in detailed NLP exam-oriented form.

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Chapter 1 (Part 2): Hugging Face Transformers and Transformer Applications

This section explains how the **Hugging Face Transformers library** makes it easy to use powerful pretrained transformer models for real-world NLP applications.

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1. The Problem Before Hugging Face

Before the Hugging Face library existed, using a new NLP model was difficult.

Whenever researchers published a new model, developers had to:

Step 1

Implement the architecture manually.

Example:

- BERT architecture
- GPT architecture
- T5 architecture

Step 2

Download pretrained weights.

Weights contain the knowledge learned during training.

Step 3

Perform preprocessing.

Example:

```
Input Text
↓
Tokenization
↓
Numerical Conversion
↓
Model Input
```

Step 4

Train the model.

Need to:

- Create datasets
- Define loss functions
- Define optimizers
- Write training loops

This process could take days or weeks.

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2. What is Hugging Face Transformers?

Hugging Face created the Transformers library to simplify all these steps.

Features

- Standardized API
- Supports hundreds of models
- Works with:
  - PyTorch
  - TensorFlow
  - JAX
- Easy fine-tuning
- Easy inference

Instead of writing hundreds of lines of code, you can use just a few lines.

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3. What is a Pipeline?

The most beginner-friendly feature in Transformers is:

Pipeline

A pipeline automatically performs:

```
Raw Text
↓
Tokenization
↓
Model Processing
↓
Prediction
```

```
↓
Output Formatting
```

You only provide text.

The pipeline does everything else.

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General Syntax

```python
from transformers import pipeline

pipe = pipeline("task-name")
```

Example:

```python
classifier = pipeline("text-classification")
```

The first time:

- Downloads model
- Downloads weights

Later:

- Uses cached version

Much faster.

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4. Text Classification

Definition

Text classification means assigning a category to text.

Examples:

Text	Class
I love this phone	Positive
Worst product ever	Negative
Sports news	Sports
Election update	Politics

Sentiment Analysis

A special type of classification.

Goal:

Determine emotional tone.

Classes:

```
POSITIVE
NEGATIVE
```

Example

Customer complaint:

```
I ordered Optimus Prime
but received Megatron.
```

Model output:

Label	Score
NEGATIVE	0.90

Meaning:

90% confidence that sentiment is negative.

Why Negative?

Customer:

- Received wrong product
- Angry
- Demanding replacement

Therefore sentiment = negative.
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Exam Definition

Text Classification

> The process of assigning predefined categories or labels to text documents.

Sentiment Analysis

> Determining whether a text expresses positive, negative, or neutral emotions.

---

5. Named Entity Recognition (NER)

What is an Entity?

Entities are real-world objects.

Examples:

- Person
- Organization
- Location
- Product

Example Sentence

```
Amazon shipped Optimus Prime to Germany.
```

Entities:

Word	Entity Type
Amazon	Organization
Optimus Prime	Product/Misc
Germany	Location

Named Entity Recognition

NER automatically finds and classifies entities.

Example Output

Entity	Category
Amazon	ORG
Germany	LOC
Bumblebee	PER

Where:

ORG

Organization

Examples:

- Google
- Amazon

- Microsoft

LOC

Location

Examples:

- India
- Germany
- Bangalore

PER

Person

Examples:

- John
- Alice
- Bumblebee

MISC

Miscellaneous entities.

Examples:

- Optimus Prime
- Megatron

Pasted text

Why is NER Important?

Applications:

Chatbots

Identify:

```
Book ticket to Bangalore
```

Location = Bangalore

Search Engines

Identify entities in queries.

Information Extraction

Extract:

- Company names
- Products
- Locations

from documents automatically.

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---

6. Tokenization

The chapter briefly introduces tokenization.

What is Tokenization?

Tokenization = Breaking text into smaller units.

Example:

```
I love NLP
```

↓

```
["I","love","NLP"]
```

These units are called:

Tokens

Subword Tokenization

Sometimes:

```
Megatron
```

becomes:

```
Mega
##tron
```

The symbol:

```
##
```

means:

"This piece belongs to the previous token."

This helps models handle rare words.

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7. Question Answering (QA)

Question Answering extracts answers from text.

Inputs

Two inputs are required:

Context

Passage containing information.

Question

What you want to know.

Example

Context:

```
I demand an exchange of Megatron.
```

Question:

```
What does the customer want?
```

Output:

```
an exchange of Megatron
```

Extractive Question Answering

The answer is directly extracted from the text.

No new sentence is generated.

Example:

Context:

```
John lives in Delhi.
```

Question:

```
Where does John live?
```

Answer:

```
Delhi
```

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Applications of Question Answering

- Customer support
- Search engines
- Digital assistants
- Legal document analysis
- Medical information retrieval

8. Summarization

Definition

Summarization converts long text into a shorter version.

Goal:

Keep important information.

Remove unnecessary details.

Example

Original:

```
Bumblebee ordered Optimus Prime.
He received Megatron instead.
```

Summary:

```
Bumblebee received the wrong action figure.
```

Types of Summarization

Extractive

Copies important sentences.

Abstractive

Creates new sentences.

Transformer models often perform:

Abstractive Summarization

because they can generate text.

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Applications

- News summaries
- Research papers
- Legal documents
- Meeting notes

---

9. Machine Translation

Definition

Converting text from one language to another.

Example:

```
English
↓
German
```

Example

Input:

```
I love NLP
```

Output:

```
Ich liebe NLP
```

(German)

Why Transformers Excel?

Attention helps learn word relationships.

Example:

English:

```
Area
```

French:

```
Zone
```

The model learns correct alignment.

Applications

- Google Translate
- International customer support
- Multilingual chatbots

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---

10. Text Generation

Definition

Generating new text based on a prompt.

Example

Prompt:

```
Once upon a time
```

Generated:

```
Once upon a time there was a dragon...
```

How GPT Works

Predict next word repeatedly.

```
Word1
↓
Predict Word2
↓
Predict Word3
```

```
↓
Predict Word4
```

This process continues until a complete response is generated.

Example from Chapter

Prompt:

```
Customer service response:
Dear Bumblebee...
```

Model generates a complete customer-service reply.

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Applications of Text Generation

ChatGPT

Conversation generation.

Email Writing

Automatic replies.

Content Creation

Blogs, articles, marketing content.

Code Generation

GitHub Copilot-style assistants.

---

11. Hugging Face Hub

The chapter mentions the Hugging Face Hub.

Think of it as:

```
GitHub
+
Model Repository
```

for machine learning models.

Users can:

- Upload models
- Download models

- Share datasets
- Fine-tune models

Thousands of pretrained transformer models are available there.

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Complete NLP Task Flow

```
Raw Text
      ↓
 Tokenization
      ↓
 Transformer Model
      ↓
 Task-Specific Head
      ↓
 Output
```

Examples:

```
Transformer
      ↓
Classification Head
      ↓
Sentiment Analysis
```

```
Transformer
      ↓
NER Head
      ↓
Entity Recognition
```

```
Transformer
      ↓
QA Head
      ↓
Question Answering
```

```
Transformer
      ↓
Generation Head
      ↓
Text Generation
```

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Exam-Oriented Summary

Important Definitions

Hugging Face Transformers

A library that provides pretrained transformer models and easy APIs for NLP tasks.

Pipeline

A high-level API that performs preprocessing, inference, and postprocessing automatically.

Text Classification

Assigning categories to text.

Named Entity Recognition (NER)

Identifying entities such as people, organizations, and locations.

Question Answering

Extracting answers from text passages.

Summarization

Generating a concise version of long text.

Machine Translation

Converting text from one language to another.

Text Generation

Producing new text based on an input prompt.

5-Mark Questions

1. What is Hugging Face Transformers?
2. Explain the Pipeline API.
3. What is Named Entity Recognition?
4. Explain Question Answering.
5. What is Text Summarization?

10-Mark Questions

1. Explain major applications of Transformer models.
2. Discuss Hugging Face Transformers architecture and benefits.
3. Compare NER, QA, Summarization, and Text Generation.
4. Explain how pipelines simplify NLP tasks.

This completes the **second part of Chapter 1: Hello Transformers** in detailed, exam-oriented form.

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The Hugging Face Ecosystem
What started with Transformers has quickly grown into a whole ecosystem consisting of many libraries and tools to accelerate your NLP and machine learning projects. The Hugging Face ecosystem consists of mainly two parts: a family of libraries and the Hub, as shown in Figure 1-9. The libraries provide the code while the Hub provides the pretrained model weights, datasets, scripts for the evaluation metrics, and more. In this section we'll have a brief look at the various components. We'll skip Transformers, as we've already discussed it and we will see a lot more of it throughout the course of the book.
Figure 1-9. An overview of the Hugging Face ecosystem
The Hugging Face Ecosystem | 15
The Hugging Face Hub
As outlined earlier, transfer learning is one of the key factors driving the success of transformers because it makes it possible to reuse pretrained models for new tasks. Consequently, it is crucial to be able to load pretrained models quickly and run experiments with them.
The Hugging Face Hub hosts over 20,000 freely available models. As shown in Figure 1-10, there are filters for tasks, frameworks, datasets, and more that are designed to help you navigate the Hub and quickly find promising candidates. As we've seen with the pipelines, loading a promising model in your code is then literally just one line of code away. This makes experimenting with a wide range of models simple, and allows you to focus on the domain-specific parts of your project.
Figure 1-10. The Models page of the Hugging Face Hub, showing filters on the left and a list of models on the right
In addition to model weights, the Hub also hosts datasets and scripts for computing metrics, which let you reproduce published results or leverage additional data for your application.
The Hub also provides model and dataset cards to document the contents of models and datasets and help you make an informed decision about whether they're the right ones for you. One of the coolest features of the Hub is that you can try out any model directly through the various task-specific interactive widgets as shown in Figure 1-11.
16 | Chapter 1: Hello Transformers
12 Rust is a high-performance programming language.
Figure 1-11. An example model card from the Hugging Face Hub: the inference widget, which allows you to interact with the model, is shown on the right
Let's continue our tour with Tokenizers.
PyTorch and TensorFlow also offer hubs of their own and are worth checking out if a particular model or dataset is not available on the Hugging Face Hub.
Hugging Face Tokenizers
Behind each of the pipeline examples that we've seen in this chapter is a tokenization step that splits the raw text into smaller pieces called tokens. We'll see how this works in detail in Chapter 2, but for now it's enough to understand that tokens may be words, parts of words, or just characters like punctuation. Transformer models are trained on numerical representations of these tokens, so getting this step right is pretty important for the whole NLP project!
Tokenizers provides many tokenization strategies and is extremely fast at tokenizing text thanks to its Rust backend.12 It also takes care of all the pre- and postprocessing steps, such as normalizing the inputs and transforming the model outputs to the required format. With Tokenizers, we can load a tokenizer in the same way we can

load pretrained model weights with Transformers.

The Hugging Face Ecosystem | 17

We need a dataset and metrics to train and evaluate models, so let's take a look at Datasets, which is in charge of that aspect.

Hugging Face Datasets

Loading, processing, and storing datasets can be a cumbersome process, especially when the datasets get too large to fit in your laptop's RAM. In addition, you usually need to implement various scripts to download the data and transform it into a standard format.

Datasets simplifies this process by providing a standard interface for thousands of datasets that can be found on the Hub. It also provides smart caching (so you don't have to redo your preprocessing each time you run your code) and avoids RAM limitations by leveraging a special mechanism called memory mapping that stores the contents of a file in virtual memory and enables multiple processes to modify a file more efficiently. The library is also interoperable with popular frameworks like Pandas and NumPy, so you don't have to leave the comfort of your favorite data wrangling tools.

Having a good dataset and powerful model is worthless, however, if you can't reliably measure the performance. Unfortunately, classic NLP metrics come with many different implementations that can vary slightly and lead to deceptive results. By providing the scripts for many metrics, Datasets helps make experiments more reproducible and the results more trustworthy.

With the Transformers, Tokenizers, and Datasets libraries we have everything we need to train our very own transformer models! However, as we'll see in Chapter 10 there are situations where we need fine-grained control over the training loop. That's where the last library of the ecosystem comes into play: Accelerate.

Hugging Face Accelerate

If you've ever had to write your own training script in PyTorch, chances are that you've had some headaches when trying to port the code that runs on your laptop to the code that runs on your organization's cluster. Accelerate adds a layer of abstraction to your normal training loops that takes care of all the custom logic necessary for the training infrastructure. This literally accelerates your workflow by simplifying the change of infrastructure when necessary.

This sums up the core components of Hugging Face's open source ecosystem. But before wrapping up this chapter, let's take a look at a few of the common challenges that come with trying to deploy transformers in the real world.

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Main Challenges with Transformers

In this chapter we've gotten a glimpse of the wide range of NLP tasks that can be tackled with transformer models. Reading the media headlines, it can sometimes sound like their capabilities are limitless. However, despite their usefulness, transformers are far from being a silver bullet. Here are a few challenges associated with them that we will explore throughout the book:

Language

NLP research is dominated by the English language. There are several models for other languages, but it is harder to find pretrained models for rare or lowresource languages. In Chapter 4, we'll explore multilingual transformers and their ability to perform zero-shot cross-lingual transfer.

Data availability

Although we can use transfer learning to dramatically reduce the amount of labeled training data our models need, it is still a lot compared to how much a

human needs to perform the task. Tackling scenarios where you have little to no labeled data is the subject of Chapter 9.
Working with long documents
Self-attention works extremely well on paragraph-long texts, but it becomes very expensive when we move to longer texts like whole documents. Approaches to mitigate this are discussed in Chapter 11.
Opacity
As with other deep learning models, transformers are to a large extent opaque. It is hard or impossible to unravel "why" a model made a certain prediction. This is an especially hard challenge when these models are deployed to make critical decisions. We'll explore some ways to probe the errors of transformer models in Chapters 2 and 4.
Bias
Transformer models are predominantly pretrained on text data from the internet. This imprints all the biases that are present in the data into the models. Making sure that these are neither racist, sexist, or worse is a challenging task. We discuss some of these issues in more detail in Chapter 10.
Although daunting, many of these challenges can be overcome. As well as in the specific chapters mentioned, we will touch on these topics in almost every chapter ahead.
Main Challenges with Transformers | 19
Conclusion
Hopefully, by now you are excited to learn how to start training and integrating these versatile models into your own applications! You've seen in this chapter that with just a few lines of code you can use state-of-the-art models for classification, named entity recognition, question answering, translation, and summarization, but this is really just the "tip of the iceberg."
In the following chapters you will learn how to adapt transformers to a wide range of use cases, such as building a text classifier, or a lightweight model for production, or even training a language model from scratch. We'll be taking a hands-on approach, which means that for every concept covered there will be accompanying code that you can run on Google Colab or your own GPU machine.
Now that we're armed with the basic concepts behind transformers, it's time to get our hands dirty with our first application: text classification. That's the topic of the next chapter!
Show moreShow less

Chapter 1 (Part 3): The Hugging Face Ecosystem, Challenges of Transformers, and Conclusion

This section explains the complete **Hugging Face Ecosystem**, its major libraries, the **Hub**, and the practical challenges faced when deploying Transformer models in real-world NLP applications.

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1. What is the Hugging Face Ecosystem?

The **Hugging Face Ecosystem** is a collection of tools, libraries, datasets, and pretrained models designed to simplify NLP and Machine Learning development.

Think of it as:

```
Hugging Face Ecosystem
        |
  ┌─────┴──────┐
Libraries      Hub
(Code)         (Resources)
```

The ecosystem has two major components:

1. Libraries

Provide code and APIs.

Examples:

- Transformers
- Tokenizers
- Datasets
- Accelerate

---

2. Hub

Provides resources such as:

- Pretrained models
- Datasets
- Evaluation metrics
- Model cards
- Dataset cards

Together they create a complete NLP development environment.

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2. The Hugging Face Hub

The Hub is one of the most important parts of the ecosystem.

You can think of it as:

```
GitHub
+
Model Zoo
+
Dataset Repository
```

for machine learning.

Why is the Hub Important?

Modern NLP depends heavily on:

Transfer Learning

Instead of training models from scratch:

```
Download Pretrained Model
        ↓
Fine-Tune
        ↓
Deploy
```

This saves:

- Time
- Money
- Computing resources

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What Does the Hub Contain?

Pretrained Models

Thousands of models are available.

Examples:

- BERT
- GPT
- RoBERTa
- T5
- DistilBERT

Datasets

Examples:

- IMDb Reviews
- SQuAD
- GLUE

- Common Crawl

Evaluation Metrics

Examples:

- Accuracy
- BLEU
- ROUGE
- F1 Score

Model Cards

Documentation describing:

- Model purpose
- Training data
- Limitations
- Ethical considerations

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---

3. Interactive Widgets

One exciting feature of the Hub is:

Interactive Inference Widgets

You can test a model directly in your browser.

Example:

```
Input:
I love NLP

Output:
Positive Sentiment
```

No coding required.

Benefits:

✅ Fast experimentation

✅ Model comparison

✅ Learning and debugging

Pasted text

4. Hugging Face Tokenizers

Before a Transformer can understand text:

Text must be converted into tokens.

This is the job of:

Tokenizers Library

Pasted text

What is Tokenization?

Tokenization means:

Breaking text into smaller pieces.

Example:

```
I love NLP
```

↓

```
["I","love","NLP"]
```

These pieces are called:

Tokens

Why Tokenization is Necessary

Computers cannot directly understand text.

Transformer models process:

```
Text
 ↓
Tokens
 ↓
Numbers
 ↓
Model
```

Without tokenization, NLP is impossible.

Features of Tokenizers

Extremely Fast

Implemented using:

Rust

which is known for:

- High performance
- Low memory usage

Handles Preprocessing

Examples:

- Lowercasing
- Cleaning text
- Normalization

Handles Postprocessing

Example:

Model output:

```
["Mega","##tron"]
```

↓

```
Megatron
```

Pasted text

---

5. Hugging Face Datasets

Training requires data.

Managing datasets manually becomes difficult when datasets are huge.

To solve this problem Hugging Face created:

Datasets Library

Pasted text

Problems Without Datasets Library

Developers must:

- Download data manually

- Clean data
- Convert formats
- Store efficiently

This becomes difficult for datasets containing millions of examples.

What Datasets Provides

Standard Dataset Interface

Load datasets using a few lines of code.

Example:

```
Python
from datasets import load_dataset
dataset = load_dataset("imdb")
```

Thousands of Datasets

Available directly from the Hub.

Examples:

- IMDb
- Wikipedia
- SQuAD
- GLUE

Smart Caching

First run:

```
Download Dataset
```

Later runs:

```
Load Cached Version
```

This saves time.

Pasted text

---

6. Memory Mapping

One important feature mentioned in the chapter is:

Memory Mapping

Large datasets often exceed RAM capacity.

Example:

```
Dataset = 100 GB
RAM = 8 GB
```

Normally impossible.

Solution

Memory mapping stores data on disk but behaves as if it is in memory.

Benefits:

- Faster access
- Lower RAM usage
- Efficient processing

This allows huge datasets to be used on normal machines.

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---

7. Dataset Compatibility

The Datasets library works with:

Pandas

Used for:

- Data analysis
- Data cleaning

NumPy

Used for:

- Numerical computation
- Matrix operations

This makes integration easy.

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8. Metrics and Reproducibility

A model is useless if performance cannot be measured.

Common NLP metrics:

Accuracy

Percentage of correct predictions.

Precision

How many predicted positives are actually positive.

Recall

How many actual positives were identified.

F1 Score

Balance between precision and recall.

BLEU

Used for machine translation.

ROUGE

Used for summarization.

The Datasets library provides implementations of these metrics.

This ensures:

✅ Reproducibility

✅ Consistency

✅ Fair comparison

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---

9. Hugging Face Accelerate

Training models on different hardware is often difficult.

Example:

Code written on:

```
Laptop
```

must later run on:

```
GPU Server
```

or

```
Multi-GPU Cluster
```

This requires substantial modifications.

What is Accelerate?

Accelerate simplifies training across different hardware platforms.

It adds an abstraction layer over training code.

```
Training Code
      ↓
Accelerate
      ↓
CPU / GPU / Multi-GPU
```

Benefits:

- Easier deployment
- Less coding
- Faster experimentation

Pasted text

Complete Hugging Face Ecosystem

```
            Hugging Face
                  │
  ┌───────────────┼──────────────┐
  │               │              │
Hub          Transformers      Tokenizers
  │               │              │
Datasets      Models         Text Processing
  │
Accelerate
  │
Training Infrastructure
```

Pasted text

---

10. Main Challenges of Transformers

Although Transformers are powerful, they have limitations.

The chapter discusses five major challenges.

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Challenge 1: Language Diversity

Most NLP research focuses on:

English language

Many languages have fewer resources.

Examples:

- Kannada
- Telugu
- Tamil
- Nepali
- Tribal languages

These are called:

Low-Resource Languages

Problems:

- Less training data
- Fewer pretrained models
- Lower accuracy

Solution

Multilingual Transformers.

Examples:

- mBERT
- XLM-R

These models support multiple languages.

Pasted text

Challenge 2: Data Availability

Humans learn from few examples.

Machines require thousands or millions.

Example:

Human:

```
5 examples
```

Machine:

```
5000 examples
```

Transfer learning reduces this requirement but does not eliminate it.

Problem:

Labeled datasets are expensive.

Pasted text

Challenge 3: Long Documents

Transformers use:

Self-Attention

Self-attention compares every word with every other word.

For N words:

Complexity:

$$O(N^2)$$

Meaning:

Doubling document length quadruples computation.

Example

Short sentence:

```
20 words
```

Easy.

Research paper:

```
10,000 words
```

Very expensive.

This is a major limitation.

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Challenge 4: Opacity (Black Box Problem)

Transformers are difficult to interpret.

Suppose model predicts:

```
Negative Review
```

Question:

```
Why?
```

Often difficult to answer.

This is called:

Model Opacity

or

Black Box Nature

Problems:

- Legal applications
- Medical applications
- Financial applications

Need explainability.

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Challenge 5: Bias

Transformer models learn from internet data.

Internet data contains:

- Cultural bias
- Gender bias
- Racial bias
- Political bias

The model may unintentionally learn these patterns.

Example

If training data contains stereotypes:

Model may reproduce stereotypes.

This creates ethical concerns.

Researchers actively work on:

- Fairness
- Debiasing
- Responsible AI

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11. Conclusion of Chapter 1

This chapter introduced the complete foundation of modern NLP.

You learned:

Core Concepts

- Encoder-Decoder Architecture
- Attention Mechanism
- Self-Attention
- Transfer Learning

Major Models

- BERT
- GPT
- Transformer Architecture

Hugging Face Tools

- Transformers
- Tokenizers
- Datasets
- Accelerate
- Hub

Applications

- Text Classification
- Sentiment Analysis

- NER
- Question Answering
- Summarization
- Translation
- Text Generation

Challenges

- Language limitations
- Lack of data
- Long documents
- Black-box behavior
- Bias

Pasted text

Very Important Exam Questions

2 Marks

1. What is the Hugging Face Hub?
2. Define tokenization.
3. What is memory mapping?
4. What is model bias?
5. What is transfer learning?

5 Marks

1. Explain the Hugging Face ecosystem.
2. Discuss the Tokenizers library.
3. Explain the Datasets library.
4. What is Accelerate?
5. Explain challenges of Transformer models.

10 Marks

1. Describe the Hugging Face ecosystem with architecture.
2. Explain the role of Hub, Tokenizers, Datasets, and Accelerate.
3. Discuss major challenges faced by Transformer models.
4. Explain how Hugging Face simplifies NLP development.

This completes the **full Chapter 1: Hello Transformers** in detailed NLP exam-oriented form.

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Chapter 2: Text Classification (Part 1)

This chapter introduces one of the most important NLP tasks: **Text Classification**. It also explains how to use the Hugging Face ecosystem to build an emotion classification model using **DistilBERT**.

Pasted text

1. What is Text Classification?

Definition

Text Classification is the process of assigning predefined categories (labels) to a text document.

Examples

Text	Category
I love this movie	Positive
Worst phone ever	Negative
Buy now and get free prizes	Spam
Meeting at 10 AM tomorrow	Not Spam

Real-World Applications

Email Spam Detection

```
Email
 ↓
Classifier
 ↓
Spam / Not Spam
```

Customer Feedback Analysis

```
Customer Review
 ↓
Classifier
 ↓
Complaint / Suggestion / Praise
```

Language Detection

```
Text
 ↓
Classifier
 ↓
English / Hindi / German
```

Social Media Monitoring

Companies analyze tweets and posts to understand customer opinions.

Example:

A company like Tesla can analyze tweets to understand whether people like a new product feature.

Pasted text

---

2. Emotion Classification

The chapter focuses on a specialized text classification problem.

Instead of:

```
Positive
Negative
```

we predict emotions.

Six Emotions in the Dataset

Label	Emotion
0	Sadness
1	Joy
2	Love
3	Anger
4	Fear
5	Surprise

Example:

```
I am extremely happy today.
```

↓

```
Joy
```

```
I feel scared about the exam.
```

↓

```
Fear
```

Pasted text

3. Why DistilBERT?

The model used in this chapter is:

DistilBERT

DistilBERT is a compressed version of BERT.

What is Knowledge Distillation?

A large model teaches a smaller model.

```
BERT
(Teacher)
    ↓
Knowledge Transfer
    ↓
DistilBERT
(Student)
```

The smaller model learns most of the knowledge from the larger model.

Advantages of DistilBERT

Smaller

Fewer parameters.

Faster

Training and inference are faster.

Cheaper

Less memory usage.

Comparable Performance

Accuracy close to BERT.

Therefore DistilBERT is commonly used in practical applications.

Pasted text

4. What is a Checkpoint?

The chapter introduces an important term:

Checkpoint

A checkpoint is:

> A saved set of pretrained model weights.

Think of it as:

```
Architecture
+
Learned Knowledge
=
Checkpoint
```

Example:

```
distilbert-base-uncased
```

is a checkpoint.

When loading a model:

```
Python
model = AutoModel.from_pretrained(
    "distilbert-base-uncased"
)
```

the weights are loaded from the checkpoint.

Pasted text

5. Typical NLP Pipeline

The chapter presents a standard workflow.

```
Raw Text
    ↓
Dataset
    ↓
Tokenizer
    ↓
Transformer
    ↓
Classifier
    ↓
Prediction
```

The three major Hugging Face libraries used are:

1. Datasets

2. Tokenizers
3. Transformers

Pasted text

6. Emotion Dataset

The dataset used is:

Emotion Dataset

This dataset contains Twitter messages.

Each tweet belongs to one of six emotions.

Example

Tweet:

```
i didnt feel humiliated
```

Label:

```
sadness
```

Pasted text

7. Hugging Face Datasets Library

To access datasets:

```python
from datasets import load_dataset
```

List Available Datasets

```python
from datasets import list_datasets

all_datasets = list_datasets()
```

This returns all datasets available on the Hub.

Examples:

- ag_news

- ai2_arc
- allegro_reviews
- emotion

Pasted text

8. Loading a Dataset

Load emotion dataset:

```
Python
from datasets import load_dataset

emotions = load_dataset("emotion")
```

Output:

```
DatasetDict
```

containing:

```
train
validation
test
```

splits.

Pasted text

9. Dataset Splits

Machine learning datasets are usually divided into three parts.

Training Set

Used to train the model.

Emotion dataset:

```
16000 examples
```

Validation Set

Used to tune hyperparameters.

```
2000 examples
```

Test Set

Used for final evaluation.

```
2000 examples
```

Why Split Data?

Without splitting:

```
Model memorizes data
```

With splitting:

```
Model generalizes
```

Pasted text

10. Dataset Object

A dataset behaves like a Python dictionary.

Access training split:

```python
train_ds = emotions["train"]
```

Length of Dataset

```python
len(train_ds)
```

Output:

```
16000
```

Access Single Example

```python
train_ds[0]
```

Output:

```python
{
 'label': 0,
```

```
 'text': 'i didnt feel humiliated'
}
```

Each row is represented as a dictionary.

Pasted text

11. Dataset Columns

Dataset contains:

```
Python
train_ds.column_names
```

Output:

```
Python
['text', 'label']
```

Text Column

Contains tweet.

Example:

```
i am feeling happy
```

Label Column

Contains emotion category.

Example:

```
joy
```

represented as an integer.

Pasted text

12. Dataset Features

```
Python
train_ds.features
```

Output:

```
Python
{
 'text': string,
 'label': ClassLabel
}
```

What is ClassLabel?

Special data type for labels.

Stores:

- Number of classes
- Label names
- Mapping between integers and labels

Example:

Integer	Emotion
0	sadness
1	joy
2	love
3	anger
4	fear
5	surprise

Pasted text

---

13. Apache Arrow

The chapter mentions:

Apache Arrow

Datasets uses Arrow internally.

Why Arrow?

Traditional Python objects consume more memory.

Arrow provides:

Columnar Storage

```
Text Column
-----------
Tweet1
Tweet2
Tweet3

Label Column
------------
0
```

```
1
2
```

Advantages

- Faster access
- Lower memory consumption
- Efficient processing

Pasted text

14. Loading Local Datasets

Not every dataset is on the Hub.

Datasets can load local files.

CSV

```Python
load_dataset(
    "csv",
    data_files="file.csv"
)
```

Text

```Python
load_dataset(
    "text",
    data_files="file.txt"
)
```

JSON

```Python
load_dataset(
    "json",
    data_files="file.jsonl"
)
```

Pasted text

15. Loading Remote Datasets

Datasets can load directly from URLs.

Example:

```
Python
load_dataset(
    "csv",
    data_files=url
)
```

Benefits:

- Automatic download
- Automatic caching

No manual downloading needed.

Pasted text

16. Converting Dataset to Pandas DataFrame

Datasets supports:

Pandas Integration

Convert:

```
Python
emotions.set_format(
    type="pandas"
)
```

Then:

```
Python
df = emotions["train"][:]
```

Result:

text	label
Tweet	0

Pasted text

17. Label Conversion

Labels are stored as integers.

Example:

```
0
```

Instead of:

```
sadness
```

Use:

```
Python
int2str()
```

to convert.

Example

Integer	Emotion
0	sadness
1	joy
2	love

This makes analysis easier.

Pasted text

Exam-Oriented Definitions

Text Classification

Assigning predefined categories to text.

Emotion Classification

Classifying text into emotions such as joy, anger, fear, and sadness.

DistilBERT

A smaller and faster version of BERT obtained through knowledge distillation.

Checkpoint

A saved set of pretrained model parameters.

Dataset Split

Division of data into training, validation, and test sets.

Apache Arrow

A memory-efficient columnar storage format used by Hugging Face Datasets.

ClassLabel

A dataset feature that maps class names to integer labels.

5-Mark Questions

1. What is text classification?
2. Explain DistilBERT.
3. What is a checkpoint?
4. Explain dataset splits.
5. What is the Hugging Face Datasets library?

10-Mark Questions

1. Explain the workflow of a text classification system.
2. Discuss the emotion dataset used in Chapter 2.
3. Explain how datasets are loaded using Hugging Face Datasets.
4. Describe the Dataset object and its important features.

This covers the **first half of Chapter 2**, up to dataset loading, inspection, and DataFrame conversion. The next section explains:

- Class distribution analysis
- Tweet length analysis
- Tokenization
- Character tokenization
- Word tokenization
- Subword tokenization (WordPiece)
- AutoTokenizer
- DistilBERT tokenizer in detail.

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Document

Chapter 2: Text Classification (Part 2)

This section covers:

1. Tokenizing the whole dataset
2. Attention Masks and Padding
3. Training a Text Classifier

4. Feature Extraction Approach
5. Hidden States
6. UMAP Visualization
7. Logistic Regression Classifier
8. Fine-Tuning DistilBERT
9. Trainer API
10. Evaluation Metrics

This is one of the most important sections for exams and interviews.
Pasted text

---

1. Tokenizing the Whole Dataset

Previously we tokenized a single sentence.

Now we need to tokenize every tweet in the dataset.

Instead of:

```python
tokenizer(text1)
tokenizer(text2)
tokenizer(text3)
```

we use:

```python
map()
```

which automatically processes all examples.

The tokenize() Function

```python
def tokenize(batch):
    return tokenizer(
        batch["text"],
        padding=True,
        truncation=True
    )
```

Here:

batch["text"]

Contains multiple tweets.

Example:

```
[
 "I am happy",
 "I am very excited today"
]
```

padding=True

Makes all sequences equal length.

---

truncation=True

Cuts sequences that exceed maximum length.

For DistilBERT:

```
Maximum Length = 512 tokens
```

Pasted text

---

2. Why Padding is Needed?

Neural networks process batches.

Suppose:

Tweet 1:

```
I am happy
```

Length:

```
3 tokens
```

Tweet 2:

```
I am extremely happy today
```

Length:

```
5 tokens
```

Problem:

```
[3 tokens]
[5 tokens]
```

cannot form a rectangular tensor.

Solution:

Add padding tokens.

```
Tweet1:
[I am happy PAD PAD]

Tweet2:
[I am extremely happy today]
```

Now both contain:

```
5 tokens
```

Pasted text

3. Special Tokens

BERT-family models use special tokens.

Token	ID	Purpose
[PAD]	0	Padding
[UNK]	100	Unknown word
[CLS]	101	Start of sequence
[SEP]	102	End of sequence
[MASK]	103	Masked language modeling

Example:

```
I love NLP
```

becomes:

```
[CLS]
I
love
NLP
[SEP]
```

Pasted text

4. Attention Mask

Padding creates another problem.

Example:

```
[CLS] I love NLP [SEP] PAD PAD PAD
```

The model should ignore PAD tokens.

Solution

Attention Mask

Example:

```
Tokens:
[CLS] I love NLP [SEP] PAD PAD

Attention Mask:
1 1 1 1 1 0 0
```

Meaning:

1

Real token

0

Ignore token

Thus the model only attends to meaningful words.

Pasted text

5. Applying Tokenization to Entire Dataset

```python
emotions_encoded = emotions.map(
    tokenize,
    batched=True
)
```

What Happens?

Original dataset:

Text	Label
Tweet	Emotion

After tokenization:

Text	Label	input_ids	attention_mask
Tweet	Emotion	Tokens	Mask

New columns are added automatically.

Output:

```python
[
 'attention_mask',
 'input_ids',
 'label',
```

```
 'text'
]
```

Pasted text

6. Training a Text Classifier

DistilBERT was originally trained for:

Masked Language Modeling

Example:

```
I love [MASK]
```

Predict:

```
NLP
```

But our task is:

```
Tweet
 ↓
Emotion
```

Therefore we must modify the architecture.

Pasted text

7. DistilBERT Architecture

The classification architecture consists of:

```
Input Text
      ↓
Tokenizer
      ↓
Embeddings
      ↓
DistilBERT Encoder
      ↓
Hidden States
      ↓
Classification Head
      ↓
Emotion Label
```

Pasted text

8. Token Encodings

After tokenization:

```
I love NLP
```

becomes:

```
[1045, 2293, 17953]
```

These are:

Token IDs

or

Input IDs

Initially represented as:

One-Hot Vectors

Example:

Word ID = 3

```
[0,0,0,1,0,0]
```

Only one position contains 1.

Hence:

One-Hot Encoding

Pasted text

9. Embeddings

One-hot vectors are huge.

Instead:

```
Token ID
     ↓
Embedding Layer
     ↓
Dense Vector
```

Example:

```
"love"
```

↓

```
[0.34, -0.76, 1.25, ...]
```

This compressed representation captures meaning.

Words with similar meanings obtain similar embeddings.

Pasted text

10. Hidden States

After passing through DistilBERT:

Each token receives a contextual representation.

Example:

```
I love NLP
```

↓

```
Hidden State Vectors
```

Each hidden state has:

```
768 dimensions
```

For DistilBERT.

Therefore:

```
Token
 ↓
768-dimensional vector
```

Pasted text

11. Feature Extraction Approach

The chapter introduces two training strategies.

Strategy 1

Feature Extraction

Freeze DistilBERT.

Use hidden states as features.

Train only a classifier.

Architecture:

```
Tweet
  ↓
DistilBERT (Frozen)
  ↓
Hidden States
  ↓
Classifier
  ↓
Emotion
```

Advantages:

✅ Fast

✅ Less computation

✅ Works without GPU

Disadvantages:

❌ Lower accuracy

Pasted text

12. Loading DistilBERT

```python
from transformers import AutoModel

model = AutoModel.from_pretrained(
    "distilbert-base-uncased"
)
```

AutoModel

Loads:

```
Pretrained DistilBERT
```

without classification head.

Only feature extraction.

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13. GPU Usage

```python
device =
torch.device(
 "cuda"
 if torch.cuda.is_available()
 else "cpu"
)
```

CUDA

NVIDIA GPU support.

Benefits:

- Faster training
- Faster inference

Pasted text

14. Last Hidden State

Input:

```
this is a test
```

Output shape:

```
Python
[1, 6, 768]
```

Meaning:

Dimension	Meaning
1	Batch size
6	Tokens
768	Hidden dimensions

Interpretation

Each token receives:

```
768 features
```

Pasted text

15. CLS Token Representation

For classification:

We usually use:

```
[CLS]
```

hidden state.

Why?

Because during training it learns a summary of the entire sentence.

Extraction:

```
Python
outputs.last_hidden_state[:,0]
```

Shape:

```
Python
[1,768]
```

Thus:

```
Whole Tweet
 ↓
768 Features
```

Pasted text

16. Extracting Hidden States

Function:

```
Python
extract_hidden_states()
```

Purpose:

```
Tweet
 ↓
DistilBERT
 ↓
CLS Vector
 ↓
Store in Dataset
```

New column:

```
hidden_state
```

is added.

Dataset becomes:

text	label	hidden_state
Tweet	Emotion	768-vector

Pasted text

17. Feature Matrix

Machine learning algorithms expect:

```
X = Features
y = Labels
```

Features

```
Python
X_train
```

Shape:

```
Python
(16000,768)
```

Meaning:

```
16000 tweets
768 features each
```

Labels

```
Python
y_train
```

Emotion IDs.

Pasted text

18. UMAP Visualization

768 dimensions are impossible to visualize.

Solution:

UMAP

(Uniform Manifold Approximation and Projection)

Purpose:

```
768D
 ↓
2D
```

for visualization.

Observed Patterns

Joy and Love

Cluster together.

Anger, Fear, Sadness

Cluster together.

Surprise

Scattered everywhere.

This suggests emotions are partially separable.

Pasted text

19. Logistic Regression Classifier

Using hidden states:

```
Python
LogisticRegression()
```

is trained.

Result:

```
Accuracy ≈ 63%
```

Pasted text

20. Baseline Classifier

Dummy classifier:

Always predicts most frequent class.

Accuracy:

```
35%
```

Comparison:

Model	Accuracy
Dummy	35%
Logistic Regression	63%

Thus DistilBERT features are useful.

Pasted text

21. Confusion Matrix

What is It?

Shows actual vs predicted classes.

Example:

Actual	Predicted
Joy	Joy
Fear	Sadness
Love	Joy

Observations

Most confusion occurs between:

- Fear ↔ Sadness
- Anger ↔ Sadness
- Love ↔ Joy

because emotions are semantically similar.

Pasted text

22. Fine-Tuning Transformers

Second strategy:

Fine-Tuning

Instead of freezing DistilBERT:

Train everything.

Architecture:

```
Tweet
 ↓
DistilBERT
(Trainable)
 ↓
Classification Head
 ↓
Emotion
```

Benefits:

✅ Much higher accuracy

Disadvantages:

❌ More GPU usage

❌ Slower training

Pasted text

23. AutoModelForSequenceClassification

Used for classification tasks.

```python
from transformers import AutoModelForSequenceClassification
```

Load:

```python
model = AutoModelForSequenceClassification.from_pretrained(
 model_ckpt,
 num_labels=6
)
```

Difference from AutoModel:

AutoModel	AutoModelForSequenceClassification
Feature extraction	Classification
No classifier head	Classifier head added

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24. Performance Metrics

Two metrics are used.

Accuracy

$$Accuracy = \frac{Correct\ Predictions}{Total\ Predictions}$$

F1 Score

Combines:

- Precision
- Recall

Formula:

$$F1 = \frac{2PR}{P + R}$$

Useful for imbalanced datasets.

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25. Trainer API

Hugging Face provides:

Trainer

to simplify training.

Handles:

- Training loop
- Evaluation
- Logging
- Saving checkpoints

Automatically.

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26. TrainingArguments

Stores hyperparameters.

Important parameters:

Parameter	Purpose
output_dir	Save model
num_train_epochs	Training cycles
learning_rate	Step size
batch_size	Samples per batch
weight_decay	Regularization

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27. Fine-Tuning Results

After training:

Metric	Value
Accuracy	92.25%
F1 Score	92.25%

Compared to:

Method	Accuracy
Feature Extraction	63%
Fine-Tuning	92%

Huge improvement.

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Feature Extraction vs Fine-Tuning

Feature Extraction	Fine-Tuning
Faster	Slower
Less GPU	More GPU
Freeze model	Train model
63% Accuracy	92% Accuracy
Simpler	Better performance

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Exam-Oriented Definitions

Padding

Adding extra tokens so all sequences have equal length.

Attention Mask

Binary vector indicating which tokens should be attended to.

Hidden State

Contextual representation produced by Transformer layers.

Feature Extraction

Using pretrained hidden states as fixed features.

Fine-Tuning

Training the entire transformer model on a downstream task.

CLS Token

Special token used to represent an entire sentence.

UMAP

Dimensionality reduction algorithm used for visualization.

Trainer API

High-level Hugging Face API for model training.

Important 5-Mark Questions

1. What is padding and why is it needed?
2. Explain attention masks.
3. What are hidden states?
4. Explain feature extraction.
5. What is fine-tuning?

Important 10-Mark Questions

1. Explain the DistilBERT text classification architecture.
2. Compare feature extraction and fine-tuning.
3. Explain the role of CLS token in classification.
4. Discuss Trainer API and TrainingArguments.
5. Explain tokenization, padding, and attention masks with examples.

This completes the **second major section of Chapter 2**, covering tokenization of the entire dataset, feature extraction, hidden states, classification, and fine-tuning.

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Chapter 2: Text Classification (Part 3 – Error Analysis, Model Saving, and Deployment)

This final section explains:

1. Error Analysis
2. Cross-Entropy Loss
3. Identifying Wrong Labels
4. Dataset Quality Improvement
5. Saving Models
6. Uploading Models to Hugging Face Hub
7. Using Fine-Tuned Models for Prediction
8. Real-World Challenges

This section is extremely important because it covers what happens **after training a model**, which is often asked in interviews and advanced NLP exams.

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1. Why Error Analysis?

Most students think:

```
Train Model
↓
Get Accuracy
↓
Done
```

But real machine learning projects do not end here.

A model with:

```
92% Accuracy
```

still makes mistakes.

The goal of error analysis is:

> Understand WHY the model makes mistakes.

Error analysis helps us:

✅ Find wrong labels

✅ Detect dataset problems

✅ Improve performance

✅ Understand model behavior

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2. What is Loss?

During training every prediction receives a score called:

Loss

Loss measures:

```
Prediction Error
```

Low Loss

Prediction is correct and confident.

Example:

```
Actual = Joy
Predicted = Joy
Confidence = 99%
```

Loss:

```
Very Small
```

High Loss

Prediction is wrong or uncertain.

Example:

```
Actual = Joy
Predicted = Sadness
```

Loss:

```
Large
```

Pasted text

3. Cross-Entropy Loss

The chapter uses:

```
Python
cross_entropy()
```

Definition

Cross-Entropy measures the difference between:

1. Actual labels
2. Predicted probabilities

Formula:

$$Loss = -\sum y_i \log(p_i)$$

Where:

- y_i = true label
- p_i = predicted probability

Example

Actual:

```
Joy
```

Model prediction:

Emotion	Probability
Joy	0.95
Sadness	0.03
Anger	0.02

Loss:

```
Very Low
```

Wrong prediction:

Emotion	Probability
Joy	0.10
Sadness	0.80
Anger	0.10

Loss:

```
Very High
```

Pasted text

---

4. Forward Pass with Labels

The chapter defines:

```python
forward_pass_with_label()
```

Purpose:

```
Input Tweet
      ↓
Model
      ↓
Prediction
      ↓
Loss Calculation
```

Returns:

```python
{
  "loss": value,
  "predicted_label": label
}
```

for every tweet.

Pasted text

5. Computing Loss for Entire Validation Set

Using:

```python
map()
```

the function is applied to all validation examples.

```python
emotions_encoded["validation"].map(...)
```

Result:

Every tweet now contains:

- Text
- Actual Label
- Predicted Label
- Loss

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6. Why Sort by Loss?

After computing losses:

```python
sort_values("loss")
```

can rank examples.

Two possibilities:

Highest Loss

Most difficult examples.

Lowest Loss

Easiest examples.

Both provide useful information.

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7. High-Loss Examples

The chapter examines tweets with the largest losses.

Example:

```
I feel betrayed...
```

Actual label:

```
Joy
```

Predicted:

```
Sadness
```

Question:

Was the model wrong?

OR

Was the label wrong?

Often:

```
"betrayed"
```

sounds much closer to:

```
Sadness
```

than:

```
Joy
```

Thus the dataset label may be incorrect.

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8. Detecting Wrong Labels

Error analysis helps identify:

Annotation Errors

Humans sometimes assign incorrect labels.

Example:

Tweet:

```
I feel betrayed.
```

Label:

```
Joy
```

Clearly suspicious.

Why This Matters

Bad labels confuse the model.

```
Wrong Labels
      ↓
Bad Learning
      ↓
Lower Accuracy
```

Fixing labels often improves performance dramatically.

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---

9. Detecting Dataset Quirks

Real datasets are messy.

Examples:

- Typos
- Emojis
- URLs
- HTML tags
- Strange punctuation

These can affect predictions.

Error analysis helps discover such problems.

Example:

```
http://badplaydate
```

appeared inside a tweet.

The model may not understand it properly.

Pasted text

10. Low-Loss Examples

Now examine:

```
Smallest Loss
```

tweets.

Examples:

```
I feel disheartened.
```

```
I feel ungrateful.
```

```
I feel broken.
```

All labeled:

```
Sadness
```

and predicted:

```
Sadness
```

with extremely high confidence.

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11. Why Analyze Easy Examples?

Many people analyze only errors.

But examining highly confident predictions is also important.

Reason:

Deep learning models sometimes exploit shortcuts.

Example:

Suppose every sadness tweet contains:

```
disheartened
```

The model may learn:

```
disheartened
↓
sadness
```

instead of truly understanding emotion.

This is called:

Shortcut Learning

or

Spurious Correlation

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12. Lessons from Error Analysis

The chapter discovered:

Observation 1

Some joy labels appear incorrect.

Observation 2

The model is very confident on sadness.

Observation 3

Certain emotions overlap naturally.

Examples:

```
Fear ↔ Sadness
Love ↔ Joy
```

These insights guide dataset improvement.

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13. Saving a Trained Model

Training a model can take:

- Minutes
- Hours
- Days

We don't want to retrain every time.

Therefore:

Save the Model

Hugging Face makes this easy.

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14. Hugging Face Hub Sharing

Upload model using:

```
Python
trainer.push_to_hub()
```

Example:

```
Python
trainer.push_to_hub(
    commit_message="Training completed!"
)
```

This uploads:

- Model weights
- Configuration
- Tokenizer
- Training information

to your Hugging Face account.

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Benefits of Sharing Models

Reproducibility

Others can reproduce results.

Collaboration

Teams can reuse models.

Deployment

Model becomes accessible online.

Community Contribution

Others can improve your work.

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15. Loading a Fine-Tuned Model

After uploading:

```
Python
classifier = pipeline(
    "text-classification",
    model=model_id
)
```

Now anyone can use your model.

Exactly like using BERT or DistilBERT.

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16. Predicting New Tweets

Example:

```
I saw a movie today and it was really good.
```

Input:

```
Tweet
```

↓

```
Fine-Tuned DistilBERT
```

↓

Output:

Emotion	Probability
Joy	Highest
Sadness	Low

Emotion	Probability
Fear	Very Low

Final prediction:

```
Joy
```

which makes sense.

Pasted text

17. Complete Text Classification Workflow

```
Raw Tweets
      ↓
Load Dataset
      ↓
Explore Data
      ↓
Tokenization
      ↓
Padding + Attention Mask
      ↓
DistilBERT
      ↓
Feature Extraction
or
Fine-Tuning
      ↓
Evaluation
      ↓
Error Analysis
      ↓
Save Model
      ↓
Deploy Model
```

This entire pipeline represents a real NLP project.

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18. Real-World Challenges Mentioned in Conclusion

The chapter concludes by discussing practical problems.

Challenge 1: Production Deployment

Boss says:

```
Deploy it today.
```

Need:

- APIs
- Cloud hosting
- Scalability

Hugging Face provides:

Inference Endpoints

for deployment.

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Challenge 2: Faster Predictions

Users want instant responses.

Solutions:

DistilBERT

Smaller model.

Model Compression

Reduce size.

Knowledge Distillation

Transfer knowledge from large model to small model.

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Challenge 3: New Tasks

Business asks:

```
Can your model also detect sarcasm?
```

or

```
Can it classify products?
```

Transformers are versatile enough to adapt to many tasks.

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Challenge 4: Multilingual Data

Not all text is English.

Example:

```
Kannada
Hindi
Tamil
French
German
```

Need multilingual transformers.

Examples:

- mBERT
- XLM-RoBERTa

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Challenge 5: No Labels

Many organizations have:

```
Huge Data
Few Labels
```

Fine-tuning becomes difficult.

Solutions:

- Transfer Learning
- Semi-Supervised Learning
- Few-Shot Learning
- Zero-Shot Learning

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Final Comparison

Feature Extraction	Fine-Tuning
Freeze Transformer	Train Transformer
Faster	Slower
Less GPU	More GPU
Easier	More Complex
63% Accuracy	92% Accuracy
Good Baseline	Best Performance

Chapter 2 Complete Summary

You learned:

Dataset Handling

- Hugging Face Datasets
- Dataset Splits
- Apache Arrow

Tokenization

- Character Tokenization
- Word Tokenization
- Subword Tokenization
- WordPiece
- AutoTokenizer

Transformer Inputs

- Input IDs
- Special Tokens
- Padding
- Attention Masks

DistilBERT

- Hidden States
- CLS Token
- Feature Extraction

Training

- Logistic Regression
- Fine-Tuning
- Trainer API
- TrainingArguments

Evaluation

- Accuracy
- F1 Score

- Confusion Matrix
- Error Analysis

Deployment

- Save Model
- Push to Hub
- Inference Pipeline

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Most Important Exam Questions (10 Marks)

1. Explain the complete text classification pipeline using DistilBERT.
2. Compare feature extraction and fine-tuning approaches.
3. Explain tokenization, padding, and attention masks with examples.
4. Discuss error analysis in transformer-based text classification.
5. Explain Trainer API and TrainingArguments in Hugging Face.
6. Describe how a fine-tuned model is saved and deployed using the Hugging Face Hub.

This completes **Chapter 2: Text Classification** in detailed exam-oriented form.

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Chapter 3: Transformer Anatomy (Part 1)

This chapter explains the **internal working of Transformers**. In Chapter 2, we learned how to use transformers. Now we learn **how they are built from scratch** and how attention actually works.

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1. Why Study Transformer Anatomy?

In practice, you can use Hugging Face models without knowing the internal architecture.

Example:

```
Python
pipeline("text-classification")
```

works without understanding self-attention.

However, understanding the architecture helps you:

✅ Debug models

✅ Design new architectures

✅ Understand limitations

✅ Crack NLP interviews

✅ Build custom transformers

The goal of this chapter is:

> Learn how a Transformer works internally.

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2. Original Transformer Architecture

The original Transformer was introduced in the paper:

"Attention Is All You Need" (2017)

The architecture follows:

```
Input Sequence
      ↓
Encoder
      ↓
Hidden Representation
      ↓
Decoder
      ↓
Output Sequence
```

Example:

```
English:
The time flies

↓ Encoder

Hidden Representation

↓ Decoder

German:
Die Zeit fliegt
```

The Transformer is based on:

Encoder–Decoder Architecture

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3. Encoder and Decoder

The Transformer contains two major parts.

Encoder

Purpose:

Convert input text into contextual representations.

Input:

```
The cat sat
```

Output:

```
Context Vectors
```

The output is often called:

- Hidden State
- Context Vector
- Contextual Embedding

Decoder

Purpose:

Generate output tokens one at a time.

Example:

```
Input:
The cat sat

Output:
Le chat est assis
```

The decoder predicts:

```
Le
↓
chat
↓
est
↓
assis
```

one token at a time.

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---

4. Components of the Transformer

The Transformer contains:

```
Input Tokens
      ↓
Token Embeddings
      ↓
Positional Embeddings
      ↓
Encoder Stack
      ↓
Decoder Stack
      ↓
Output Tokens
```

Important components:

1. Token Embeddings
2. Positional Embeddings
3. Self-Attention
4. Multi-Head Attention
5. Feed Forward Network
6. Layer Normalization
7. Skip Connections

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5. Why Positional Embeddings?

Attention does not know word order.

Example:

```
Dog bites man
```

and

```
Man bites dog
```

contain the same words.

Without positions:

The model cannot distinguish them.

Solution

Add position information.

Example:

Word	Position
Dog	1

Word	Position
bites	2
man	3

This information is stored in:

Positional Embeddings

Final Input:

```
Token Embedding
+
Position Embedding
=
Transformer Input
```

Pasted text

6. Encoder Stack

The encoder is not a single layer.

It consists of multiple identical blocks.

Example:

```
Input
 ↓
Encoder Block 1
 ↓
Encoder Block 2
 ↓
Encoder Block 3
 ↓
...
 ↓
Output
```

In BERT:

```
12 Encoder Layers
```

In BERT Large:

```
24 Encoder Layers
```

Each layer refines the representation.

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7. Encoder Layer Structure

Each encoder layer contains:

```
Input
 ↓
Multi-Head Self-Attention
 ↓
Add & Normalize
 ↓
Feed Forward Network
 ↓
Add & Normalize
 ↓
Output
```

Two important sublayers:

Self-Attention

Learns relationships between words.

Feed Forward Network

Processes each token independently.

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8. What is Self-Attention?

Self-attention is the heart of the Transformer.

Basic Idea

Every word looks at every other word.

Example:

```
The animal didn't cross the street
because it was tired.
```

Question:

What does:

```
it
```

refer to?

Self-attention learns:

```
it → animal
```

without manually programming grammar rules.

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9. Formal Definition of Self-Attention

Suppose we have:

$$x_1, x_2, x_3, \ldots, x_n$$

These are token embeddings.

Self-attention creates:

$$x'_1, x'_2, x'_3, \ldots, x'_n$$

called:

Contextualized Embeddings

Formula:

$$x'_i = \sum_{j=1}^{n} w_{ji} x_j$$

Meaning:

New embedding = weighted combination of all embeddings.

Where:

$$w_{ji}$$

represents:

Attention Weight

or

Importance Score

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---

10. Contextualized Embeddings

Traditional word embeddings:

```
Apple
```

always have the same vector.

Problem:

Apple can mean:

Fruit

```
I ate an apple.
```

or

Company

```
Apple released a new iPhone.
```

Self-attention solves this.

The embedding changes according to context.

These dynamic representations are called:

Contextualized Embeddings

Pasted text

11. Example: "Flies"

Consider:

Sentence 1

```
Time flies like an arrow.
```

Here:

```
flies = verb
```

Sentence 2

```
Fruit flies like a banana.
```

Here:

```
flies = insect
```

Self-attention creates different embeddings for:

```
flies
```

in the two contexts.

This is one of the major breakthroughs of Transformers.

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12. Scaled Dot-Product Attention

The Transformer uses:

Scaled Dot-Product Attention

This is the mathematical mechanism behind self-attention.

There are four steps.

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Step 1: Create Query, Key, Value

Every token embedding is transformed into:

```
Embedding
   ↓
Query (Q)
Key (K)
Value (V)
```

Each token gets:

- Query Vector
- Key Vector
- Value Vector

Example:

```
time
```

↓

```
Qtime
Ktime
Vtime
```

Pasted text

13. Understanding Query, Key, Value

This is the most commonly asked interview topic.

Supermarket Analogy

Imagine you have a shopping list.

The item you want:

```
Milk
```

acts like:

Query

Shelf labels:

```
Milk
Bread
Eggs
```

act like:

Keys

Actual products:

```
Milk packet
Bread loaf
Egg tray
```

act like:

Values

Process:

```
Query
 ↓
Compare with Keys
 ↓
Find Match
 ↓
Retrieve Value
```

Exactly how self-attention works.

Pasted text

14. Step 2: Compute Attention Scores

The model measures similarity between:

```
Query
```

and

```
Key
```

using:

Dot Product

Formula:

$$Score = Q \cdot K$$

Large score:

```
High Similarity
```

Small score:

```
Low Similarity
```

Pasted text

15. Attention Score Matrix

Suppose sentence:

```
time flies like arrow
```

contains:

```
4 tokens
```

Attention produces:

$$4 \times 4$$

matrix.

Example:

	time	flies	like	arrow
time	•	•	•	•
flies	•	•	•	•
like	•	•	•	•
arrow	•	•	•	•

Each cell contains:

```
Attention Score
```

Pasted text

16. Step 3: Scaling

Dot products can become very large.

Large values make training unstable.

Therefore divide by:

$$\sqrt{d_k}$$

where:

$$d_k$$

= dimension of key vectors.

Formula:

$$Score = \frac{QK^T}{\sqrt{d_k}}$$

This is the:

Scaled

part of Scaled Dot Product Attention.

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17. Step 4: Softmax

Convert scores into probabilities.

Formula:

$$Softmax(x_i) = \frac{e^{x_i}}{\sum_j e^{x_j}}$$

Properties:

✅ Values between 0 and 1

✅ Sum equals 1

Example:

Scores:

```
2
1
0
```

↓

Softmax:

```
0.67
0.24
0.09
```

These become:

Attention Weights

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18. Step 5: Compute Output

Finally:

$$Output = AttentionWeights \times Values$$

or

$$Attention(Q, K, V) = Softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

This is the most important Transformer equation.

Memorize it for exams.

$$\boxed{Attention(Q, K, V) = Softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)V}$$

Pasted text

Complete Self-Attention Pipeline

```
Input Embeddings
        ↓
Create Q, K, V
        ↓
Compute QKᵀ
        ↓
Scale by √dk
        ↓
Softmax
        ↓
Attention Weights
        ↓
Multiply by V
        ↓
Contextualized Embeddings
```

Pasted text

Important Exam Questions

2 Marks

1. What is self-attention?
2. Define contextualized embedding.
3. What are Query, Key, and Value vectors?
4. Why is scaling used in attention?

5 Marks

1. Explain the Transformer architecture.
2. Explain positional embeddings.
3. Describe self-attention with an example.
4. Explain contextualized embeddings.

10 Marks

1. Derive the scaled dot-product attention equation.
2. Explain encoder architecture in detail.
3. Discuss the role of Query, Key, and Value vectors.
4. Explain how self-attention resolves ambiguity in language.

This completes **Chapter 3 Part 1**, covering:

- Transformer architecture
- Encoder–Decoder structure
- Encoder layers
- Self-attention
- Query-Key-Value mechanism
- Scaled Dot-Product Attention

The next part covers:

- Multi-Head Attention
- Attention Heads
- Feed Forward Networks
- Skip Connections
- Layer Normalization
- Full Encoder Construction
- Decoder Architecture in detail.

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Chapter 3: Transformer Anatomy (Part 2)

This section completes the **Transformer Encoder**, explains **Feed-Forward Networks, Layer Normalization, Skip Connections, Positional Embeddings, Classification Heads, and the Decoder Architecture**. These topics are among the most frequently asked in NLP interviews and university exams.
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---

1. Feed-Forward Layer (FFN)

After the Self-Attention layer, every Transformer encoder contains a:

Feed-Forward Network (FFN)

Structure:

```
Attention Output
        ↓
Linear Layer 1
        ↓
GELU Activation
        ↓
Linear Layer 2
        ↓
Dropout
        ↓
Output
```

Unlike self-attention, FFN processes each token independently.
Pasted text

Why Do We Need FFN?

Self-attention learns:

```
Word Relationships
```

Example:

```
animal ↔ it
```

But attention alone cannot perform complex nonlinear transformations.

The FFN:

- Learns patterns
- Stores knowledge

- Performs feature extraction
- Increases model capacity

Most of the model's memorization power is believed to reside in FFN layers.

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2. Position-Wise Feed-Forward Network

The FFN is often called:

Position-Wise Feed-Forward Network

Why?

Because each token is processed independently.

Example:

Sentence:

```
I love NLP
```

Embeddings:

```
Token1 → FFN
Token2 → FFN
Token3 → FFN
```

No interaction occurs between tokens inside FFN.

Token interaction happens only in:

Self-Attention

Pasted text

3. FFN Architecture

Typical structure:

$$FFN(x) = W_2(GELU(W_1x + b_1)) + b_2$$

Where:

- W_1 = first weight matrix
- W_2 = second weight matrix
- GELU = activation function

Hidden Size Expansion

Rule used in most Transformers:

$$Intermediate\ Size = 4 \times Hidden\ Size$$

Example:

BERT:

```
Hidden Size = 768
```

Intermediate size:

```
3072
```

because:

$$768 \times 4 = 3072$$

Pasted text

---

4. GELU Activation Function

Transformers usually use:

GELU (Gaussian Error Linear Unit)

instead of:

- ReLU
- Sigmoid
- Tanh

Why GELU?

Advantages:

✅ Smooth

✅ Better gradients

✅ Better performance in Transformers

Used in:

- BERT
- GPT
- RoBERTa
- DistilBERT

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5. Layer Normalization

Training deep networks is difficult because activations can become unstable.

Solution:

Layer Normalization

Purpose:

Normalize each hidden representation.

After normalization:

$$Mean = 0$$

$$Variance = 1$$

Benefits:

✅ Stable training

✅ Faster convergence

✅ Better gradient flow

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Example

Before normalization:

```
[100, 250, -75]
```

Large variance.

After LayerNorm:

```
[-0.8, 1.2, -0.4]
```

Much more stable.

Pasted text

6. Skip Connections (Residual Connections)

Another key Transformer component:

Skip Connection

Idea:

Pass input directly to output.

Instead of:

```
Input
 ↓
Layer
 ↓
Output
```

we use:

```
Input
 ↘
  +
 ↗
Layer
```

Mathematically:

$$Output = x + Layer(x)$$

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Why Skip Connections?

Deep networks suffer from:

Vanishing Gradients

Early layers stop learning.

Skip connections help gradients flow backward easily.

Benefits:

✅ Easier optimization

✅ Better training

✅ Deeper models possible

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7. Placement of Layer Normalization

The chapter discusses two designs.

Post-Layer Normalization

Original Transformer (2017)

```
Attention
 ↓
Add
 ↓
LayerNorm
```

Problem:

Training instability.

Requires:

Learning Rate Warm-Up

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Pre-Layer Normalization

Modern Transformers

```
LayerNorm
 ↓
Attention
 ↓
Add
```

Benefits:

✅ More stable

✅ Easier training

✅ No warm-up usually required

Most modern architectures use this approach.

Pasted text

8. Complete Encoder Layer

Now combine everything:

```
Input
 ↓
LayerNorm
 ↓
Multi-Head Attention
 ↓
Add (Skip Connection)
 ↓
LayerNorm
 ↓
Feed Forward Network
 ↓
Add (Skip Connection)
 ↓
Output
```

This is one complete:

Transformer Encoder Layer

Pasted text

9. Positional Embeddings

Self-attention is:

Permutation Invariant

Meaning:

```
Dog bites man
```

and

```
Man bites dog
```

look similar.

The model doesn't know word order.

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Solution

Add position information.

Every token receives:

```
Token Embedding
+
Position Embedding
```

Example:

Sentence:

```
I love NLP
```

Positions:

Token	Position
I	0
love	1
NLP	2

Each position has its own learned vector.

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Final Input Representation

$$Embedding = TokenEmbedding + PositionEmbedding$$

This combined vector enters the encoder.

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10. Types of Positional Embeddings

The chapter discusses three approaches.

A. Learnable Positional Embeddings

Most common.

Each position has trainable parameters.

Example:

```
Position 0 → vector
Position 1 → vector
Position 2 → vector
```

Learned during training.

Used by:

- BERT
- GPT

Pasted text

B. Absolute Positional Encoding

Uses mathematical functions.

Example:

$$sin(x), cos(x)$$

Advantages:

- No extra parameters

- Works well with limited data

Used in original Transformer.

Pasted text

C. Relative Positional Encoding

Encodes:

```
Distance Between Tokens
```

instead of absolute positions.

Example:

```
cat is 2 positions away from dog
```

Used by models like:

DeBERTa

Advantages:

Better handling of long-range dependencies.

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11. Full Transformer Encoder

Combining everything:

```
Input IDs
      ↓
Token Embeddings
      ↓
Position Embeddings
      ↓
Encoder Layer 1
      ↓
Encoder Layer 2
      ↓
Encoder Layer 3
      ↓
...
      ↓
Encoder Layer N
      ↓
Contextualized Representations
```

Output shape:

```
[Batch Size,
 Sequence Length,
 Hidden Size]
```

Example:

```
[1,5,768]
```

Meaning:

- 1 sentence
- 5 tokens
- 768-dimensional representation per token

Pasted text

12. Classification Head

Transformers are usually divided into:

```
Transformer Body
+
Task Head
```

Pasted text

Transformer Body

Produces hidden states.

Classification Head

Produces class predictions.

Example:

```
Tweet
 ↓
Transformer
 ↓
CLS Token
 ↓
Linear Layer
 ↓
Emotion Label
```

Pasted text

Why Use CLS Token?

BERT introduces:

```
[CLS]
```

at the beginning.

Example:

```
[CLS] I love NLP
```

The hidden state of `[CLS]` learns to summarize the entire sentence.

For classification:

```
CLS Hidden State
      ↓
Classifier
      ↓
Prediction
```

Pasted text

13. Decoder Architecture

The decoder is more complex than the encoder.

Contains:

1. Masked Multi-Head Self-Attention

2. Encoder–Decoder Attention

3. Feed Forward Network

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Decoder Structure

```
Input Tokens
      ↓
Masked Self-Attention
      ↓
Add & Norm
      ↓
Encoder-Decoder Attention
      ↓
Add & Norm
      ↓
Feed Forward
      ↓
Add & Norm
      ↓
Output
```

Pasted text

14. Why Masking is Needed?

Suppose decoder is generating:

```
I love NLP
```

While predicting:

```
love
```

the model should NOT see:

```
NLP
```

because that would be cheating.

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Causal Mask

Mask matrix:

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 \end{bmatrix}$$

Meaning:

Token can attend only to:

- itself
- previous tokens

Never future tokens.

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Example

Token 3:

```
NLP
```

can attend to:

```
I
love
NLP
```

but not future words.

This is called:

Causal Attention

or

Autoregressive Attention

Pasted text

15. Encoder–Decoder Attention

Unique to encoder-decoder models.

Purpose:

Connect source language with target language.

Example:

English:

```
The cat
```

German:

```
Die Katze
```

Decoder asks:

```
Which source word should I focus on?
```

The encoder provides information through:

Encoder–Decoder Attention

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Exam Analogy

The book gives a classroom analogy:

Decoder

Student taking exam.

Encoder

Student who already has the answer sheet.

Decoder sends:

```
Query
```

Encoder searches:

```
Matching Key
```

Returns:

```
Value
```

This is exactly how Encoder–Decoder Attention works.

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Encoder vs Decoder

Encoder	Decoder
Bidirectional	Left-to-right
Self-attention	Masked self-attention
Used in BERT	Used in GPT
Understands text	Generates text
Sees all tokens	Sees only past tokens

Pasted text

Complete Transformer Formula Flow

```
Input IDs
    ↓
Token Embeddings
    ↓
Position Embeddings
    ↓
Multi-Head Attention
    ↓
Add & LayerNorm
    ↓
Feed Forward Network
    ↓
Add & LayerNorm
    ↓
Encoder Output
```

Decoder:

```
Previous Tokens
      ↓
Masked Attention
      ↓
Encoder-Decoder Attention
      ↓
Feed Forward
      ↓
Next Token Prediction
```

Pasted text

Very Important Exam Questions

2 Marks

1. What is a feed-forward network?
2. Define layer normalization.
3. What is a skip connection?
4. What is a positional embedding?
5. What is causal masking?

5 Marks

1. Explain position-wise feed-forward networks.
2. Compare pre-layer normalization and post-layer normalization.
3. Explain positional embeddings with examples.
4. Why are skip connections needed?

10 Marks

1. Explain the complete Transformer encoder architecture.
2. Discuss feed-forward networks in Transformers.
3. Explain positional embeddings and their types.
4. Describe the Transformer decoder architecture.
5. Explain masked self-attention and causal masking.

This completes **Chapter 3: Transformer Anatomy**, covering:

- Self-Attention
- Scaled Dot-Product Attention
- Multi-Head Attention
- Feed-Forward Networks
- Layer Normalization
- Skip Connections
- Positional Embeddings
- Classification Heads
- Decoder Architecture
- Causal Masking
- Encoder–Decoder Attention.

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Pasted text.txt
Document

Chapter 3: Transformer Anatomy (Part 3)

Meet the Transformers – The Transformer Family Tree

After understanding the internal architecture of Transformers, the book introduces the **major Transformer models** that evolved from the original Transformer architecture.

Almost every modern NLP model belongs to one of three categories:

```
Transformer Models
        |
  ┌─────┼─────┐
  |     |     |
Encoder Decoder Encoder-Decoder
```

These three branches form the **Transformer Tree of Life**.

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1. The Transformer Family Tree

After the success of the original Transformer (2017), researchers started creating improved versions by:

- Using larger datasets
- Using new training objectives
- Improving architectures
- Reducing model size
- Increasing multilingual support

Although hundreds of models exist today, almost all belong to one of:

Encoder Models

(BERT family)

Decoder Models

(GPT family)

Encoder-Decoder Models

(T5, BART family)

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Overview

Architecture	Purpose	Example
Encoder	Understanding text (NLU)	BERT
Decoder	Generating text (NLG)	GPT
Encoder-Decoder	Understanding + Generation	T5

---

2. Encoder Branch

Encoder-only models focus on:

Natural Language Understanding (NLU)

Tasks:

- Text Classification
- Sentiment Analysis
- Named Entity Recognition
- Question Answering

The most famous encoder model is:

BERT

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---

3. BERT

Full Form

Bidirectional Encoder Representations from Transformers

Main Idea

BERT reads text:

```
Left ← Word → Right
```

simultaneously.

This is called:

Bidirectional Context Understanding

Pretraining Objectives

A. Masked Language Modeling (MLM)

Example:

```
I love [MASK]
```

Predict:

```
NLP
```

B. Next Sentence Prediction (NSP)

Example:

Sentence 1:

```
I love NLP.
```

Sentence 2:

```
Transformers are powerful.
```

Model predicts:

```
Does Sentence 2 follow Sentence 1?
```

This helps learn relationships between sentences.

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Advantages of BERT

✅ Strong contextual understanding

✅ State-of-the-art NLU

✅ Excellent for classification tasks

Applications

- Sentiment Analysis

- Chatbots
- Search Engines
- Question Answering

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4. DistilBERT

Problem:

BERT is large.

Issues:

- Slow
- Memory intensive
- Difficult deployment

Solution:

DistilBERT

A compressed version of BERT.

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Knowledge Distillation

```
Teacher Model
(BERT)
      ↓
Knowledge Transfer
      ↓
Student Model
(DistilBERT)
```

The student learns from the teacher.

Performance

DistilBERT achieves:

97% of BERT performance

while using:

40% less memory

and being:

60% faster

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Advantages

✅ Smaller

✅ Faster

✅ Production friendly

5. RoBERTa

Researchers discovered BERT could be improved.

Result:

RoBERTa

(Robustly Optimized BERT)

Pasted text

Improvements Over BERT

More Training Data

Larger Batch Sizes

Longer Training

Removed NSP Task

```
BERT
MLM + NSP

RoBERTa
MLM only
```

Result

Better accuracy than BERT on many benchmarks.

Pasted text

6. XLM

Cross-Lingual Language Model

Designed for:

Multilingual NLP

Supports multiple languages simultaneously.

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New Training Objective

Translation Language Modeling (TLM)

Example:

English:

```
I love NLP
```

French:

```
J'aime le NLP
```

Model learns relationships across languages.

Applications

- Multilingual Classification
- Translation
- Cross-Language Transfer

Pasted text

---

7. XLM-RoBERTa

Combines:

```
XLM
+
RoBERTa
```

↓

XLM-R

Pasted text

Training Data

Trained on:

2.5 Terabytes

of multilingual text.

Collected from:

Common Crawl

Benefits

- Better multilingual understanding
- Excellent low-resource language performance

Examples:

- Kannada
- Telugu
- Nepali
- Swahili

Pasted text

8. ALBERT

A Lite BERT

Goal:

Reduce parameters without reducing performance.

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Three Main Improvements

1. Smaller Embeddings

Embedding size separated from hidden size.

2. Parameter Sharing

All layers reuse parameters.

3. Sentence Order Prediction

Replaces NSP.

Example:

Original:

```
Sentence A
Sentence B
```

Model predicts:

```
Correct Order?
```

instead of simply:

```
Do they belong together?
```

Result

Larger models can be trained using fewer parameters.

Pasted text

9. ELECTRA

One weakness of MLM:

Only masked tokens contribute to learning.

Example:

```
I love [MASK]
```

Only one token is trained.

Pasted text

ELECTRA Solution

Uses:

Generator

Predicts masked tokens.

Discriminator

Detects which tokens were replaced.

```
Input
 ↓
Generator
 ↓
Modified Text
 ↓
Discriminator
 ↓
Real or Fake?
```

Benefit

Every token participates in learning.

Training becomes:

30× More Efficient

Pasted text

---

10. DeBERTa

Decoding-Enhanced BERT

Introduces:

Disentangled Attention

Pasted text

Key Idea

Represent each token using:

Content Vector

Stores meaning.

Position Vector

Stores location.

```
Token
 ↓
Content Vector
+
Position Vector
```

Advantages

Better modeling of nearby words.

Superior language understanding.

Achievement

First model to exceed human baseline on:

SuperGLUE Benchmark

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Encoder Model Comparison

Model	Key Idea
BERT	MLM + NSP
DistilBERT	Smaller BERT
RoBERTa	Better BERT training
XLM	Multilingual
XLM-R	Large multilingual
ALBERT	Parameter sharing
ELECTRA	Real/Fake token detection
DeBERTa	Content + Position separation

Pasted text

11. Decoder Branch

Decoder-only models focus on:

Text Generation

Examples:

- ChatGPT
- Story generation
- Email writing
- Code generation

The most famous family:

GPT

Pasted text

12. GPT

Generative Pretrained Transformer

Uses:

Autoregressive Training

Predict next word.

Example:

```
I love
```

Predict:

```
NLP
```

Architecture:

```
Decoder Only
```

No encoder.

Pasted text

Applications

- Text Generation
- Completion
- Dialogue Systems

13. GPT-2

Larger version of GPT.

Trained on much larger data.

Capabilities:

- Long coherent paragraphs
- Better reasoning
- Better text generation

Pasted text

14. CTRL

Problem:

GPT generates text but offers little control.

Solution:

CTRL

Conditional Transformer Language Model

Pasted text

Control Tokens

Example:

```
[NEWS]
```

Generate news article.

```
[RECIPE]
```

Generate cooking instructions.

This provides style control.

Pasted text

15. GPT-3

Massive scale increase.

Parameters:

175 Billion

Pasted text

Special Capability

Few-Shot Learning

Provide only a few examples.

Example:

```
English → Python
```

After seeing examples:

Model performs new tasks.

Without retraining.

Achievement

Generated highly realistic text.

Pasted text

---

16. GPT-Neo and GPT-J

Created by:

EleutherAI

Open-source alternatives to GPT-3.

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Model Sizes

GPT-Neo:

- 1.3B
- 2.7B

GPT-J:

- 6B

parameters.

Competitive with smaller GPT-3 variants.

Pasted text

Decoder Model Comparison

Model	Contribution
GPT	First major decoder model
GPT-2	Larger generation model
CTRL	Controllable generation
GPT-3	Few-shot learning
GPT-Neo	Open-source GPT
GPT-J	Larger open-source GPT

Pasted text

17. Encoder–Decoder Branch

Combines:

```
Encoder
+
Decoder
```

Used for:

Sequence-to-Sequence Tasks

Examples:

- Translation
- Summarization
- Text Generation

Pasted text

18. T5

Text-to-Text Transfer Transformer

Key Idea:

Everything becomes:

Text → Text

Pasted text

Examples

Classification:

```
Input:
I love NLP

Output:
positive
```

Translation:

```
Input:
Hello

Output:
Bonjour
```

Summarization:

```
Input:
Long article

Output:
Short summary
```

All tasks share one framework.

Pasted text

19. BART

Combines ideas from:

```
BERT
+
GPT
```

↓

BART

Pasted text

Training Process

Corrupt text:

```
I love NLP
```

↓

```
I [MASK] NLP
```

↓

Model reconstructs original text.

Applications

- Summarization
- Translation
- Text Generation
- NLU

Pasted text

20. M2M-100

Many-to-Many Translation

Translates directly between:

100 Languages

Pasted text

Example:

```
Kannada
  ↓
French
```

without first translating through English.

Major breakthrough in multilingual translation.

Pasted text

21. BigBird

Problem:

Self-attention complexity:

$$O(n^2)$$

Long documents become expensive.

Solution:

Sparse Attention

Used by BigBird.

Pasted text

Context Length

Standard BERT:

```
512 Tokens
```

BigBird:

```
4096 Tokens
```

Applications:

- Research papers
- Legal documents
- Books
- Long summarization

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Final Architecture Summary

Architecture	Models
Encoder	BERT, RoBERTa, ALBERT, ELECTRA, DeBERTa
Decoder	GPT, GPT-2, GPT-3, CTRL, GPT-Neo
Encoder-Decoder	T5, BART, M2M-100

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Most Important Exam Questions

2 Marks

1. What is BERT?
2. What is DistilBERT?
3. Define GPT.
4. What is T5?
5. What is BigBird?

5 Marks

1. Explain BERT and its pretraining objectives.
2. Compare BERT and RoBERTa.
3. Explain ELECTRA architecture.
4. Describe GPT-3 and few-shot learning.
5. Explain T5 and the text-to-text framework.

10 Marks

1. Discuss the Transformer family tree in detail.
2. Compare encoder, decoder, and encoder-decoder architectures.
3. Explain major encoder-only Transformer models.
4. Explain major decoder-only Transformer models.
5. Discuss T5, BART, M2M-100, and BigBird.

This completes **Chapter 3: Transformer Anatomy**, including the final section **"Meet the Transformers"**, which surveys the major Transformer architectures used in modern NLP.

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